Analytical Chemistry || Cumulative Exam || October 17. 2025

Instructions: Answer the following questions. Please place answers in proper numerical order in the blue book. This exam is meant to challenge you so do your best.

Part I – Short Answer

1. Define accuracy and precision. Describe a scenario where measurements are precise but not accurate.
2. Explain the main advantage of internal standards vs. external calibration. Provide an analytical situation where it is essential to use internal standards.
3. Explain how signal-to-noise ratio (S/N) affects the detection limit of an analytical method.

Part II – Data and Problem Solving

4. A 0.4250 g sample of impure NaCl is dissolved and precipitated as AgCl. The dried AgCl weighs 0.8120 g.
   a) Write the balanced chemical equation.
   b) Calculate the % NaCl in the sample. (Molar masses: NaCl = 58.44, AgCl = 143.32 g/mol)

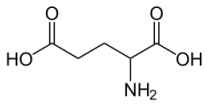
5. You titrate 25.00 mL of 0.100 M acetic acid with 0.100 M NaOH.
   a) Calculate the pH at the equivalence point.
   b) If the pK_a of acetic acid is 4.76, what is the pH halfway to equivalence?

6. You observe a molecular ion peak at *m/z* = 180 and a strong fragment at *m/z* = 165.
   a) Propose a possible explanation for the loss observed.
   b) What type of ionization (EI, ESI, MALDI) would most likely produce such fragmentation and why?

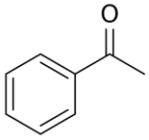
Part III – Concept Integration

7. You determine the slope of a calibration line for absorbance to be 0.105 ± 0.002 and the intercept (blank) 0.010 ± 0.005. If a sample gives A = 0.546, show how you would calculate [X] and its uncertainty. Just show the equations and approach you would use with justification and do not calculate the values.

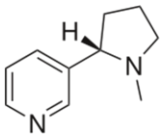
8. Calculate the pI value for glutamate. The pK_a values are 2.1 (α COOH), 4.1, 9.5 (α NH_3^+). Draw the proper structure for glutamate when the pH of a solution matches its pI. What useful separation technique relies on separation by pI?



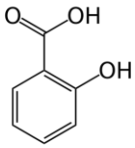
9. Consider the following compounds: acetophenone, salicylic acid (pK_a= 2.97) and nicotine (pK_{a1} (aromatic) = 3.15 and pK_{a2} = 7.85). A table of their retention behavior (k') is shown below for a C8-bonded silica HPLC column. The mobile phase contained 70/30 (v/v%) water/methanol.



Acetophenone



Nicotine



Salicylic Acid

		pH 3	**pH 5**	**pH 7**
Acetophenone		4.21	4.28	4.37
Salicylic acid		2.97	0.65	0.62
Nicotine		0.00	0.13	3.11

a) Draw the dominant species at each pH value.

b) Explain the pH-dependence for the retention behavior on this LC column.

Equation Sheet

$$\frac{U_R}{R} = \sqrt{\left(\frac{U_1}{R_1}\right)^2 + \left(\frac{U_2}{R_2}\right)^2 + \left(\frac{U_3}{R_3}\right)^2 + \cdots}$$

$$U_R = \sqrt{(U_1)^2 + (U_2)^2 + (U_3)^2 + \cdots}$$

$$pH = pK_a + \log\frac{base\ form}{acid\ form}$$

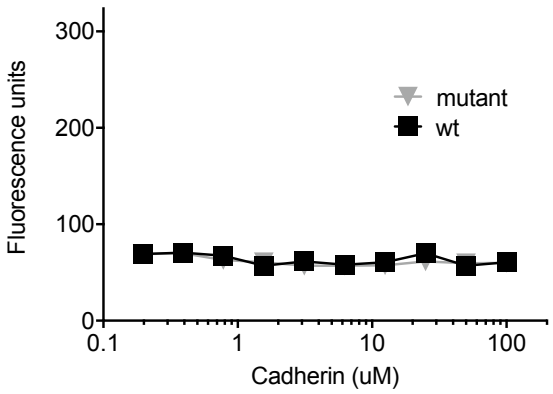
$$A = \epsilon bC$$

Isoelectric pH uses the appropriate surrounding K_a values. $pI = \frac{pK_{s1} + pK_{s2}}{2}$

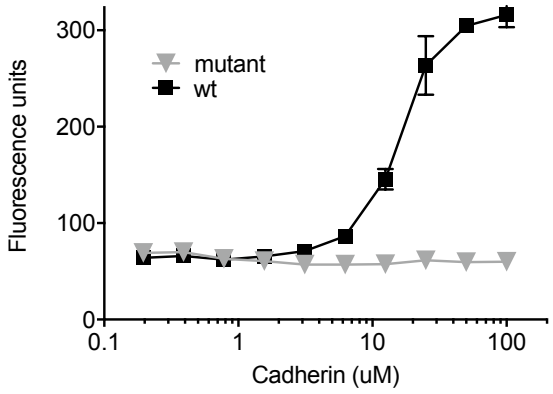
$$k' = \frac{t_r - t_m}{t_m}$$

Figure 1

A) pH 10.4



B) pH 7.4



Refer to Figure 1 for the following questions

1. An ELISA experiment was performed to determine the importance of the cationic residues at the N terminus of fibrin in its binding to cadherin. Fluorescent labeled Cadherin was titrated into wild type (wt) or Fibrin with the cationic residues each mutated to alanine (mutant). wt Fibrin peptide has an isoelectric point of 10.37, and mutant fibrin has an isoelectric point of 10.6. Which of the following is the most likely reason that binding was not observed in Figure 1A) (20 pts)

   A) The proteins stuck to the wells of the 96 well plate
   B) The fibrin precipitated out of solution
   C) The wrong protein was used in the experiment

2. By visual inspection, what is the equilibrium constant of Fibrin peptide binding to cadherin? (20 pts)

   A) 17 μM
   B) 100 nM
   C) 0.1 nM

3. Thrombin is a serine protease known to target Fibrin. How does Thrombin activate Fibrin?(20 pts)

   A) It phosphorylates Serine 53 within the Aα chains
   B) It removes the phosphate group on Serine the available amount of thiamine in the cell
   C) It cleaves the peptide bonds of fibrinogen

4. In an analytical ultracentrifuge equilibrium experiment, 50 μM Cadherin was combined with 0.24 μM wt-Fibrin. At the end of the experiment, it was estimated that 25% of Fibrin was unbound and 75% Fibrin was bound to Cadherin. Use the equation to <u>calculate the equilibrium constant</u> of the Fibrin peptide + Cadherin interaction. (40 pts)

   K = [A][B] / [AB] = [free fibrin][free cadherin] / [fibrin+cadherin complex]